context

- our data are represented with $X^{m \times n}$

$$X = \begin{pmatrix} x_{11} & x_{12} & \dots & x_{1n} \\ x_{21} & x_{22} & \dots & x_{2n} \\ \vdots & \vdots & \dots & \vdots \\ x_{m1} & x_{m2} & \dots & x_{mn} \end{pmatrix} \qquad x_i = \begin{pmatrix} x_{1i} \\ x_{2i} \\ \vdots \\ x_{mi} \end{pmatrix} \quad X = (x_1, x_2, \dots, x_n)$$

- data are in a n-dimensional space with coordinates $x_1, x_2, \dots, x_n$ (features)
- we seek a new system of orthogonal coordinates $z_1, z_2, \dots, z_p$ (principal components) with $p \leq n$ to represent orginal data such that
  1. it better captures the variance of data $\Rightarrow$ if $p < n$ the information loss is "minimal"
  2. each new coordinate z_i is a linear combination of all the original coordinates

$$\begin{aligned} z_1 &= \phi_{11}x_1 + \phi_{21}x_2 + \dots + \phi_{n1}x_n \\ z_2 &= \phi_{12}x_1 + \phi_{22}x_2 + \dots + \phi_{n2}x_n \\ &\vdots \\ z_p &= \phi_{1p}x_1 + \phi_{2p}x_2 + \dots + \phi_{np}x_n \end{aligned}$$

$$(x_1, x_2, \dots, x_n) \Rightarrow (z_1, z_2, \dots, z_p)$$

  3. thay can be ranked in order of "importance"

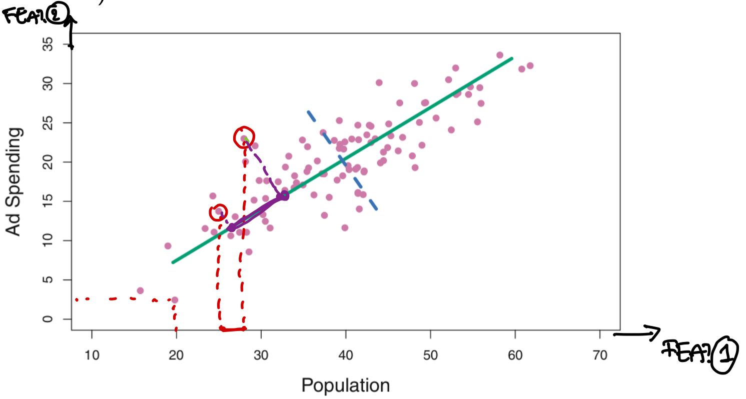
PROPORTION OF VARIANCE EXPRESSED WITH RESPECT TO THE OVERALL VARIANCE geometric interpretation

COORDINATES ARE PROJECTIONS OF POINTS OVER THE AXES

"Principal components are a sequence of projections of the data, mutually uncorrelated and ordered in variance."
('The Elements of Statistical Learning' page 534)

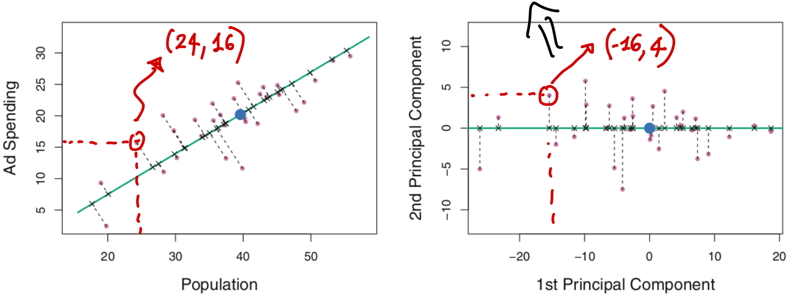
→ PCs ORTHOGONAL → THEY PROVIDE COMPLETELY DIFFERENT INFO

example: population size (feature 1 or x_1) and ad spending (feature 2 or x_2) over 100 cities (individuals)



geometric interpretation

$-16 = z_1 = \phi_{11}x_1 + \phi_{21}x_2 = \phi_{11}\cdot 24 + \phi_{21}\cdot 16$



$$\begin{aligned} z_1 &= \phi_{11}x_1 + \phi_{21}x_2 \\ z_2 &= \phi_{12}x_1 + \phi_{22}x_2 \end{aligned} \quad \Rightarrow \quad \begin{aligned} z_1 &= \phi_{11}\texttt{pop} + \phi_{21}\texttt{ad} \\ z_2 &= \phi_{12}\texttt{pop} + \phi_{22}\texttt{ad} \end{aligned}$$

coordinates (z_1, z_2) (Principal Component scores) determine the position of individuals in the new system!

$\begin{pmatrix} \phi_{11} \\ \phi_{21} \end{pmatrix}, \begin{pmatrix} \phi_{12} \\ \phi_{22} \end{pmatrix}$: loadings vectors associated to 1st and 2nd Principal Components

$$z_1 = \overset{\phi_{11}}{0.839} \times (\texttt{pop} - \overline{\texttt{pop}}) + \overset{\phi_{21}}{0.544} \times (\texttt{ad} - \overline{\texttt{ad}})$$

mathematical interpretation

we sequentially determine new coordinates (directions) maximizing the overall variance of data

AVERAGE

- center data by columns as $x_i^{centered} = x_i - \bar{x}_i \Rightarrow \bar{x}_i^{centered} = 0$
- determine the first Principal Component by solving

$$\max_{\phi_{11},\phi_{21},\dots,\phi_{n1}} \quad \frac{1}{m}\sum_{i=1,m}\sum_{j=1,n}(\phi_{j1}x_{ij} - \underset{\overset{\shortparallel}{0}}{\bar{x}_j})^2 = \frac{1}{m}\sum_{i=1,m}\sum_{j=1,n}(\phi_{j1}x_{ij})^2$$

$$\text{subject to} \qquad \sum_{j=1,n}\phi_{j1}^2 = 1$$

ϕ DETERMINED BY MAXIMIZING THE AVERAGE VARIANCE IN DATA

TO HAVE A SOLUTION OF THE PROBLEM (AVOID $\|\phi\| \to \infty$)

- determine the second Principal Component by solving

$$\max_{\phi_{12},\phi_{22},\dots,\phi_{n2}} \quad \frac{1}{m}\sum_{i=1,m}\sum_{j=1,n}(\phi_{j2}x_{ij})^2$$

$$\text{subject to} \qquad \sum_{j=1,n}\phi_{j2}^2 = 1$$

$$\sum_{j=1,n}\phi_{j1}\phi_{j2} = 0 \to \phi_1^T\phi_2 = 0$$

$\Downarrow$

ORTHOGONALITY

- ...

covariance interpretation

$cov(x_i, x_j) =$
$= E[(x_i - \bar{x}_i)(x_j - \bar{x}_j)]$

- center data by columns as $x_i^{centered} = x_i - \bar{x}_i \Rightarrow \bar{x}_i^{centered} = 0$
- construct the covariance matrix as SQUARE $n \times n$ MATRIX, SYMMETRIC

$$C = \begin{pmatrix} cov(x_1, x_1) & cov(x_1, x_2) & \dots & cov(x_1, x_n) \\ cov(x_2, x_1) & cov(x_2, x_2) & \dots & cov(x_2, x_n) \\ \vdots & \vdots & \dots & \vdots \\ cov(x_n, x_1) & cov(x_n, x_2) & \dots & cov(x_n, x_n) \end{pmatrix}$$

C symmetric, positive semidefinite

$$\Rightarrow \begin{cases} n \text{ orthogonal eigenvectors } v_1, v_2, \dots, v_n & \text{(SYMMETRIC)} \\ n \text{ nonnegative eigenvalues } \lambda_1 \geq \lambda_2 \geq \dots \geq \lambda_n \geq 0 & \text{(POSITIVE SEMIDEFINITE)} \end{cases}$$

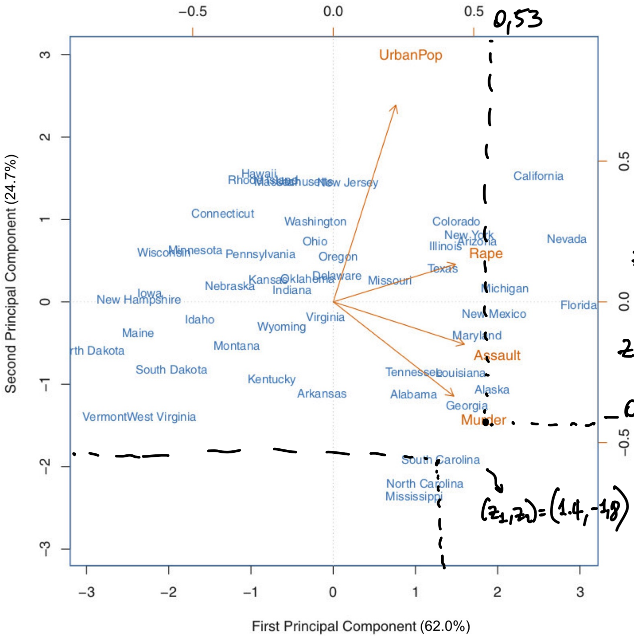
- eigenvectors represent the Principal Components LOADING VECTORS

$$v_1 = \begin{pmatrix} \phi_{11} \\ \phi_{21} \\ \vdots \\ \phi_{n1} \end{pmatrix}, v_2 = \begin{pmatrix} \phi_{12} \\ \phi_{22} \\ \vdots \\ \phi_{n2} \end{pmatrix}, \dots, v_n = \begin{pmatrix} \phi_{1n} \\ \phi_{2n} \\ \vdots \\ \phi_{nn} \end{pmatrix}$$

proportion of variance explained (PVE) by Principal Component i

$$\frac{\lambda_i}{\sum_{j=1,n} \lambda_j}$$

biplot



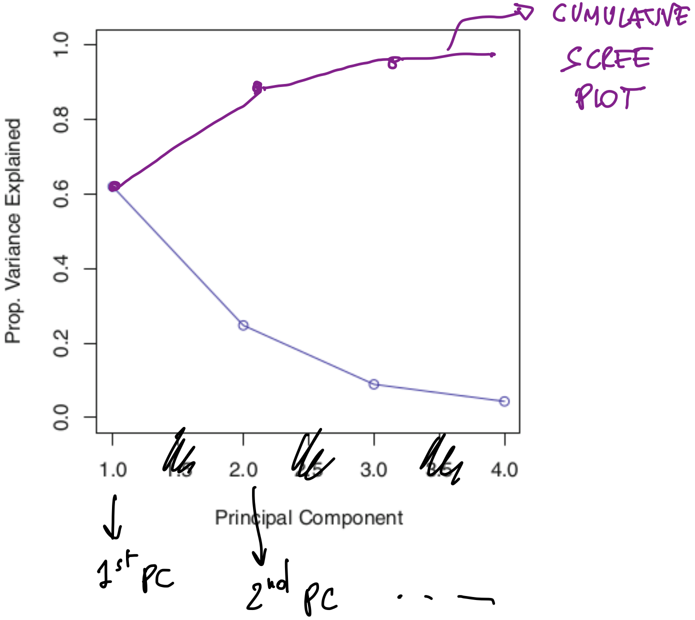
example:
Assault, Murder, Rape and Urban Population measured on 50 US states

$Z_1 = 0{,}53$ MURDER $+ 0{,}58$ ASSAULT
$+ 0{,}27$ POP $+ 0{,}54$ RAPE

$Z_2 = -0{,}41$ MURDER $- 0{,}18$ ASSAULT
$+ 0{,}87$ POP $+ 0{,}16$ RAPE $+$

	PC1	PC2
Murder	0.5358995	−0.4181809
Assault	0.5831836	−0.1879856
UrbanPop	0.2781909	0.8728062
Rape	0.5434321	0.1673186

scree plot



exercise 2.3